

Load Interconnection System Impact Study Report

For Queue Project Q22-2265

Masspower 138 kV

February 23, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2265. The Interconnection Customer proposes a 87.3 MW large load (data center) facility interconnecting at the Masspower 138 kV substation in the Virginia area. The request was received on December 28, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 427 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$824,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sewell'S Pt – Yamato 138kV	Diamond Springs – Hospers 138kV L	103.1	346	12.0
Wheelabrator Portsmouth – Reeves Ave 138kV L	Sewell'S Pt – Yamato 138kV L	102.2	336	5.6
Thole St – Reeves Ave 138kV	Yamato Substation – Sewell'S Point 138kV line	100.9	275	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sewell'S Pt – Yamato 138kV	Reconductor line	\$647,000	100% (\$647,000)
Wheelabrator Portsmouth – Reeves Ave 138kV	Reconductor line	\$1,018,000	5% (\$55,000)
Reeves Ave – Thole Street 138kV Line	Reconductor line	\$2,245,000	5% (\$122,000)
	Total allocated to project		\$824,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.